Understanding Linux drop_caches

1. Why is drop_caches necessary?

Generally, Linux manages its memory exceptionally well. If another large program requires RAM, Linux automatically frees up this Cache memory and allocates it to that program.

However, under certain special circumstances, developers or system administrators want to clear this Cache forcefully (manually). For example:

- To test the actual performance of any software or code (without the influence of Cache).
- When RAM appears to be completely full unnecessarily in the system.

2. How to use drop_caches? (Commands)

In Linux, there is a file located at `/proc/sys/vm/drop_caches`. We can clear the Cache by writing specific numbers into this file.

Note:

To run these commands, you must have **root** (Administrator) privileges. Therefore, you need to use `sudo` or `su`.

There are 3 types of methods available:

Method 1: To clear PageCache only (Option 1)

This clears the temporary data of files.

```
sudo sync; echo 1 > /proc/sys/vm/drop_caches
```

Method 2: To clear Dentries and Inodes (Option 2)

This clears the directory structure (metadata) information of files and folders.

```
sudo sync; echo 2 > /proc/sys/vm/drop_caches
```

Method 3: To clear PageCache, Dentries, and Inodes altogether (Option 3)

This completely clears both of the above-mentioned caches (the entire Cache). This is the most commonly used method.

```
sudo sync; echo 3 > /proc/sys/vm/drop_caches
```

3. Why do we use sync in the command?

You might have noticed the presence of `sudo sync;` in the commands given above. `sync` is an important command. It tells Linux to safely write all the pending data that is not yet saved in RAM (Dirty Pages) onto the hard disk. If you drop the Cache directly without running `sync`, there is a risk of losing unsaved data.

4. Advantages and Disadvantages

Advantages:

- Accurate benchmark results are obtained during testing.
- Space is temporarily freed up in RAM (you can see the difference by running the `free -m` command).

Disadvantages / Warnings:

- **Performance drops:** Immediately after clearing the Cache, the system becomes a bit slow. This is because the next time any file needs to be opened, Linux has to read it from the hard disk all over again from scratch.
- **Should not be used daily:** This should not be run minute-by-minute using automated scripts on any production server. Doing so damages the natural memory management of Linux.

Conclusion:

In Linux, `drop_caches` is an excellent tool, but it should be used only when necessary (during testing or troubleshooting) and not for day-to-day operations.

To state the difference in just a single line:

- **Page Cache:** The actual data inside the file (Content).
- **Inode Cache:** The size, permissions, and location details of that file.
- **Dentry Cache:** The folder path and the file name where that file is located.